

Basic Principles of Medicine 1
Module: Musculoskeletal System | Lecture 20

Histology of Cartilage Pre-read Material

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Recommended Reading

Histology – A Text and Atlas

Pawlina 9th Edition

Chapter 7: Cartilage

[Cartilage | Histology: A Text and Atlas: With Correlated Cell and Molecular Biology, 9e | Premium Basic Sciences | Health Library \(lwwhealthlibrary.com\)](http://www.healthlibrary.com)

[Cartilage | Histology: A Text and Atlas: With Correlated Cell and Molecular Biology, 9e | Premium Basic Sciences | Health Library \(oclc.org\)](http://oclc.org)

Objectives

List the 3 types of cartilage	SOM.MK.I.BPM1.2.MSK.3.HCB.1014
Describe the histological components specific to cartilage	SOM.MK.I.BPM1.2.MSK.3.HCB.1015
Describe the various components of extracellular matrix of cartilage in general.	SOM.MK.I.BPM1.2.MSK.3.HCB.1016
Describe the microanatomical features of chondroblasts.	SOM.MK.I.BPM1.2.MSK.3.HCB.1017
Describe the microanatomical features of chondrocytes.	SOM.MK.I.BPM1.2.MSK.3.HCB.1018
Give the function(s) of the perichondrium.	SOM.MK.I.BPM1.2.MSK.3.HCB.1019
Give the location of perichondrium.	SOM.MK.I.BPM1.2.MSK.3.HCB.1020
Describe the microscopic structure of the perichondrium.	SOM.MK.I.BPM1.2.MSK.3.HCB.1021
Discuss the two types of growth seen in cartilage in relation to the perichondrium and isogenous groups.	SOM.MK.I.BPM1.2.MSK.3.HCB.1022
Identify the location of capsular, territorial and inter-territorial matrices.	SOM.MK.I.BPM1.2.MSK.3.HCB.1023
Make a comparison table differentiating the three types of cartilage.	SOM.MK.I.BPM1.2.MSK.3.HCB.1024

Cartilage

Overview

- Specialized type of **connective tissue**:
 - Unique cells and fiber population and ground substance
 - Avascular
 - 3 types
 - ❖ Hyaline
 - ❖ Elastic
 - ❖ Fibrocartilage
- **Components**
 - **5% cells**
 - Chondroblasts
 - Chondrocytes
 - *(Chondroclasts – for cartilage remodeling or removal)
 - **95% extracellular matrix (ECM) - Ground substance - Fibers**
 - solid and firm but pliable – accounts for its resilience
 - Allows diffusion of substances (large ratio of glycosaminoglycans to type II collagen fibers)

ECM components

■ Ground substance

- **Glycosaminoglycans (GAGs)** e.g., hyaluronan, chondroitin-4-sulphate, chondroitin-6-sulphate, keratan sulphate
- **Proteoglycan** monomers (aggrecan) and aggregates
- **Multiadhesive glycoproteins**

■ Predominant **fiber** type varies based on cartilage type

- **Hyaline cartilage** – Type II Collagen
- **Elastic cartilage** – Type II collagen + Elastic
- **Fibrocartilage** – Type I + Type II collagen

Cartilage Functions

- Confers shape, flexibility & elasticity
- Shock absorber (resistance to compression)
- Tensile strength
- Model for long bone formation

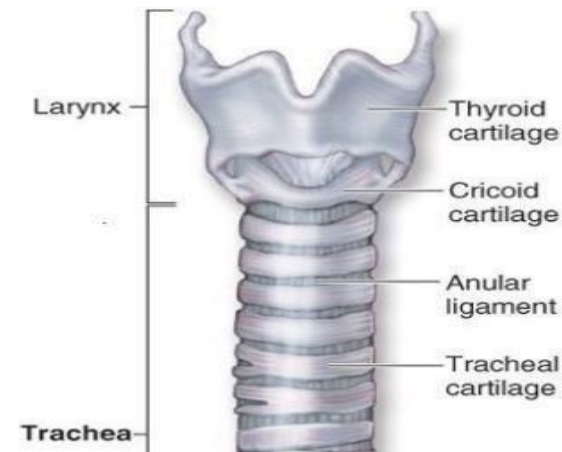
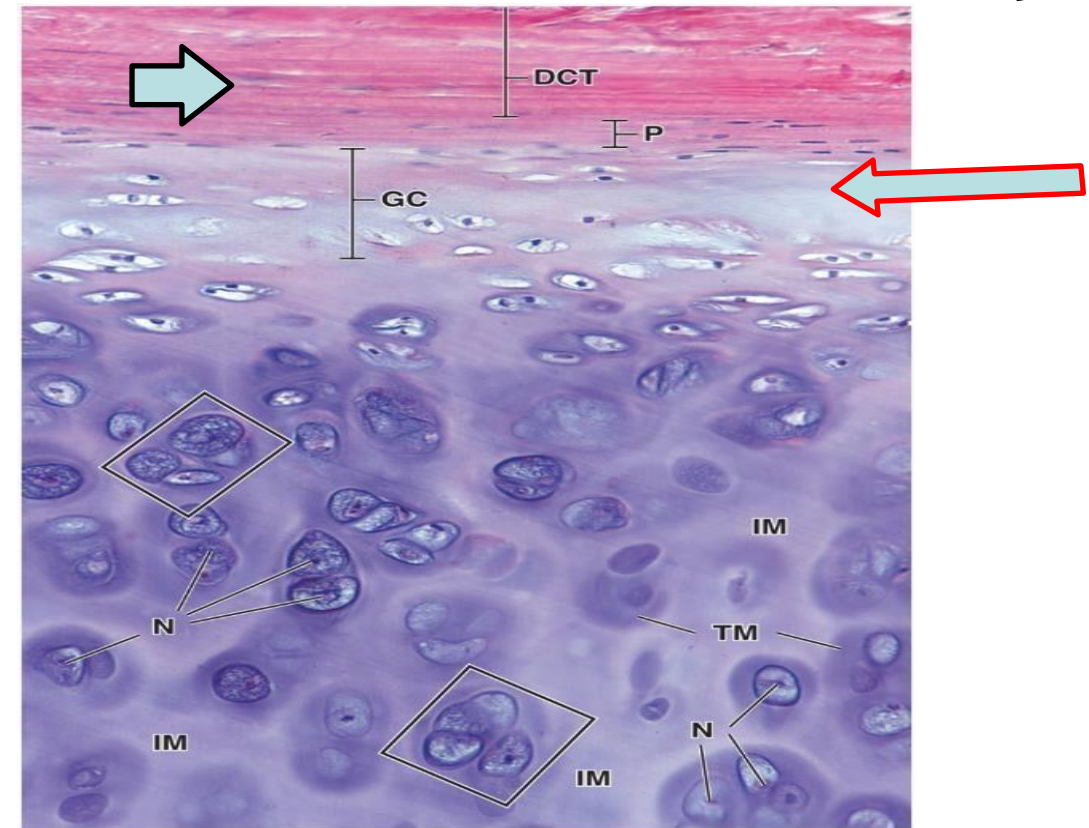
Perichondrium (P)

Connective tissue (dense irregular) covering entire cartilage (**except** articular and epiphyseal plate cartilage, and fibrocartilage)

Made up of two layers:

1. **Outer Fibrous Layer**
 - contains fibroblasts (synthesize collagen type I and matrix)
2. **Inner Cellular (Chondrogenic) Layer**
 - contains chondrogenic cells (**source of new cartilage cells**)

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Perichondrium:

Outer Fibrous layer (FL):

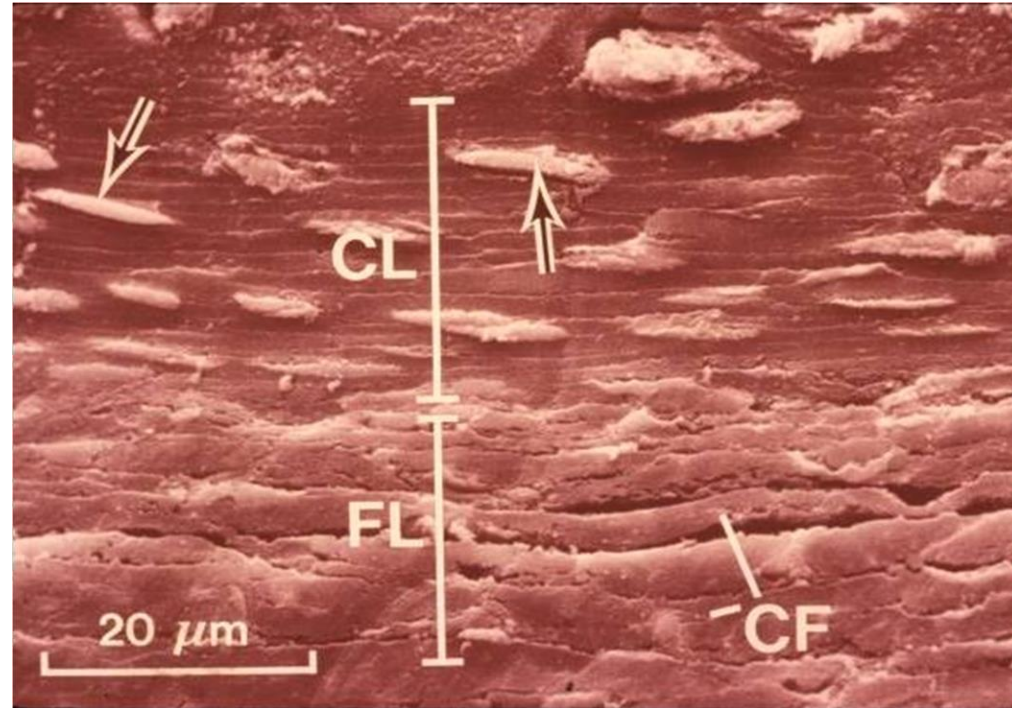
1. Contains fibroblasts
2. Type I collagen
3. Blood vessels

Inner Cellular layer (CL):

- Chondrogenic cells (Source of chondroblasts)

Cartilage is avascular !!

Nutrients are supplied by vessels from perichondrium by diffusion, **Or** from the synovial fluid at the articular surfaces.



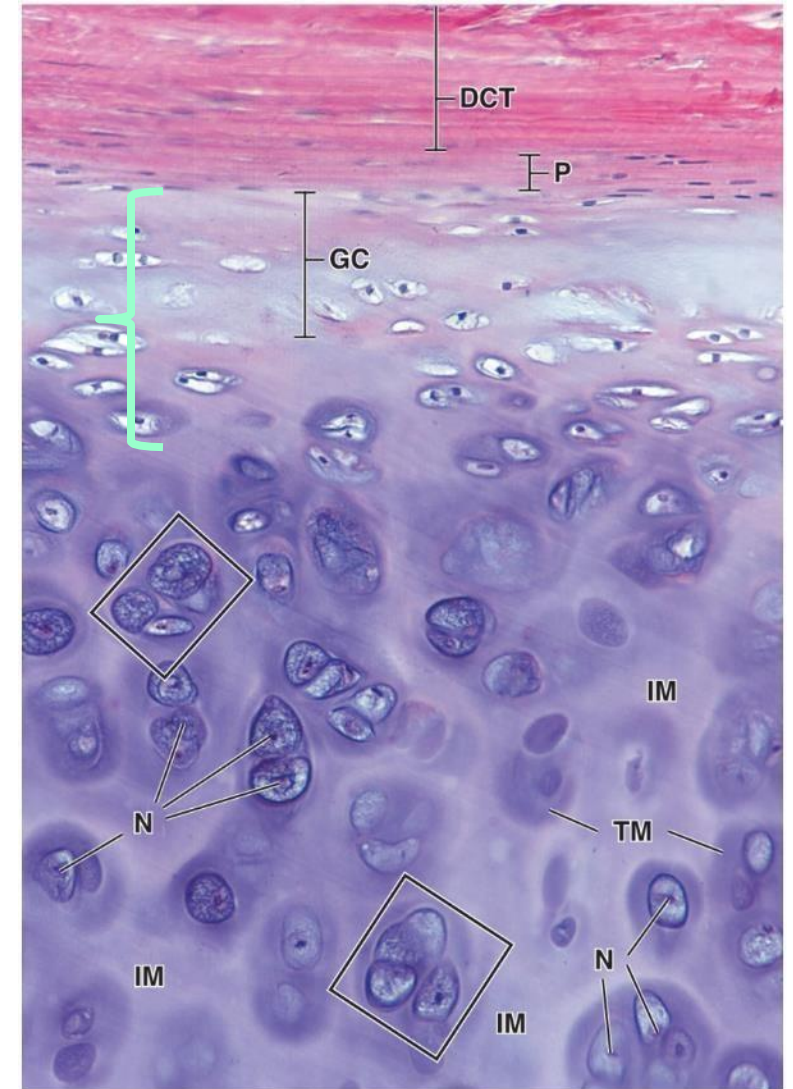
FL – Fibrous layer
CL – Chondrogenic layer
CF – Collagen fibers (type I)

Cartilage Cells

**Cells constitute only ~5% of total cartilage mass*

Chondroblasts

- Chondrogenic cells give rise to chondroblasts
- Location: just below chondrogenic layer of perichondrium (P)
- Chondroblasts are ovoid and positioned so that their longitudinal axes lie parallel to the cartilage surface
- The space occupied by a chondroblast is called a lacuna
- They synthesize cartilage matrix
- Responsible for external growth of cartilage (Appositional growth)



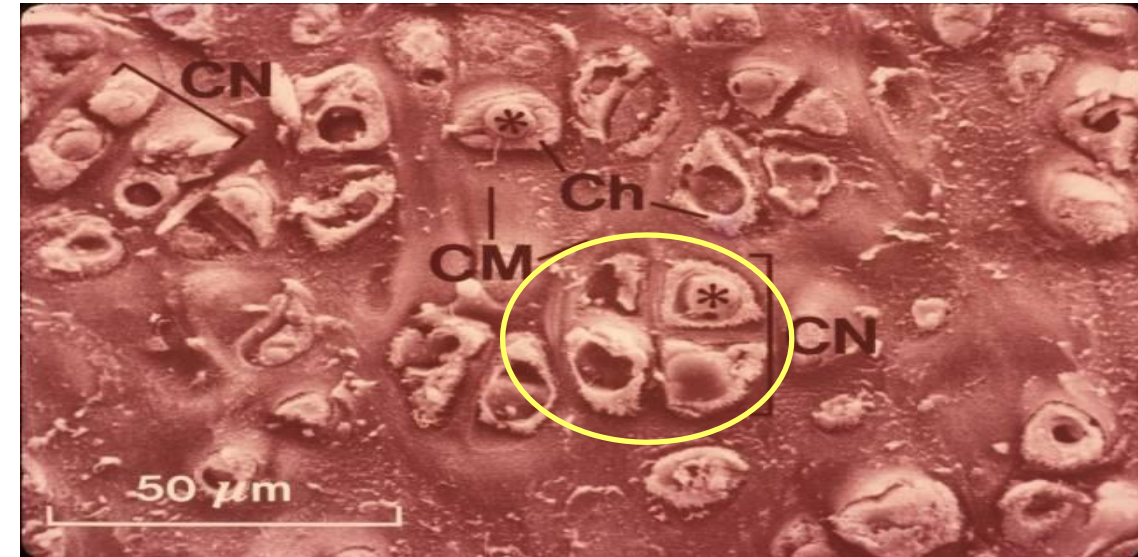
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Chondrocytes

- Matured chondroblasts located in the lacunae and surrounded by matrix
- Located deeper to the chondroblasts and are more spherical
- Divide by mitosis to form groups of 4-8 chondrocytes, referred to as **isogenous groups** or **cell nest (CN)**
 - suggestive of internal growth (Interstitial growth)

Chondroclasts

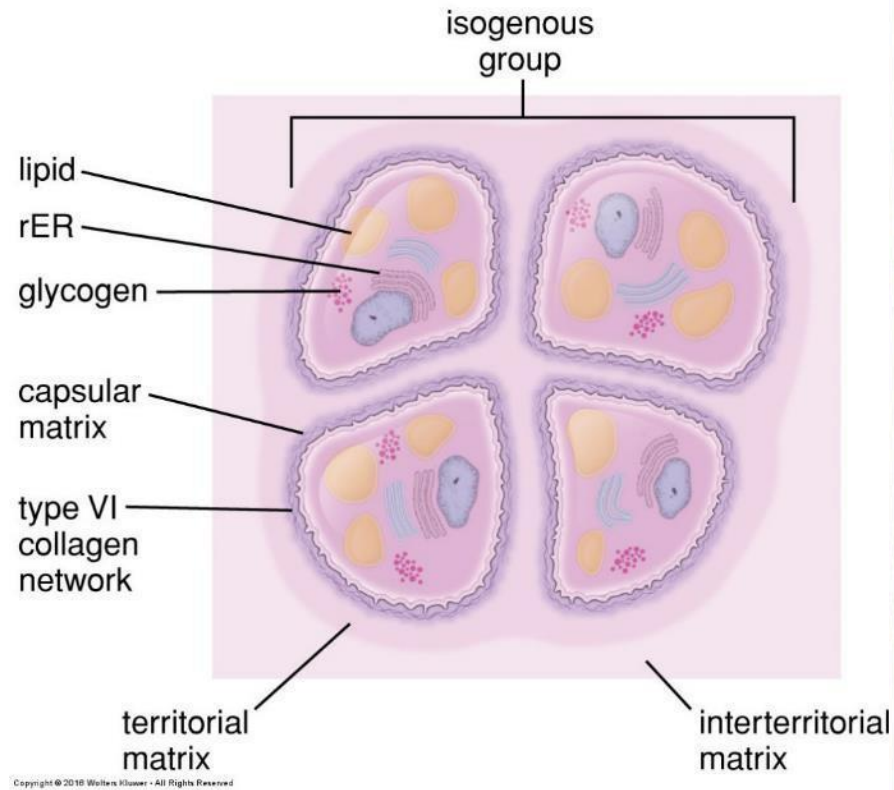
- originate from monocytes
- help to remodel the cartilage



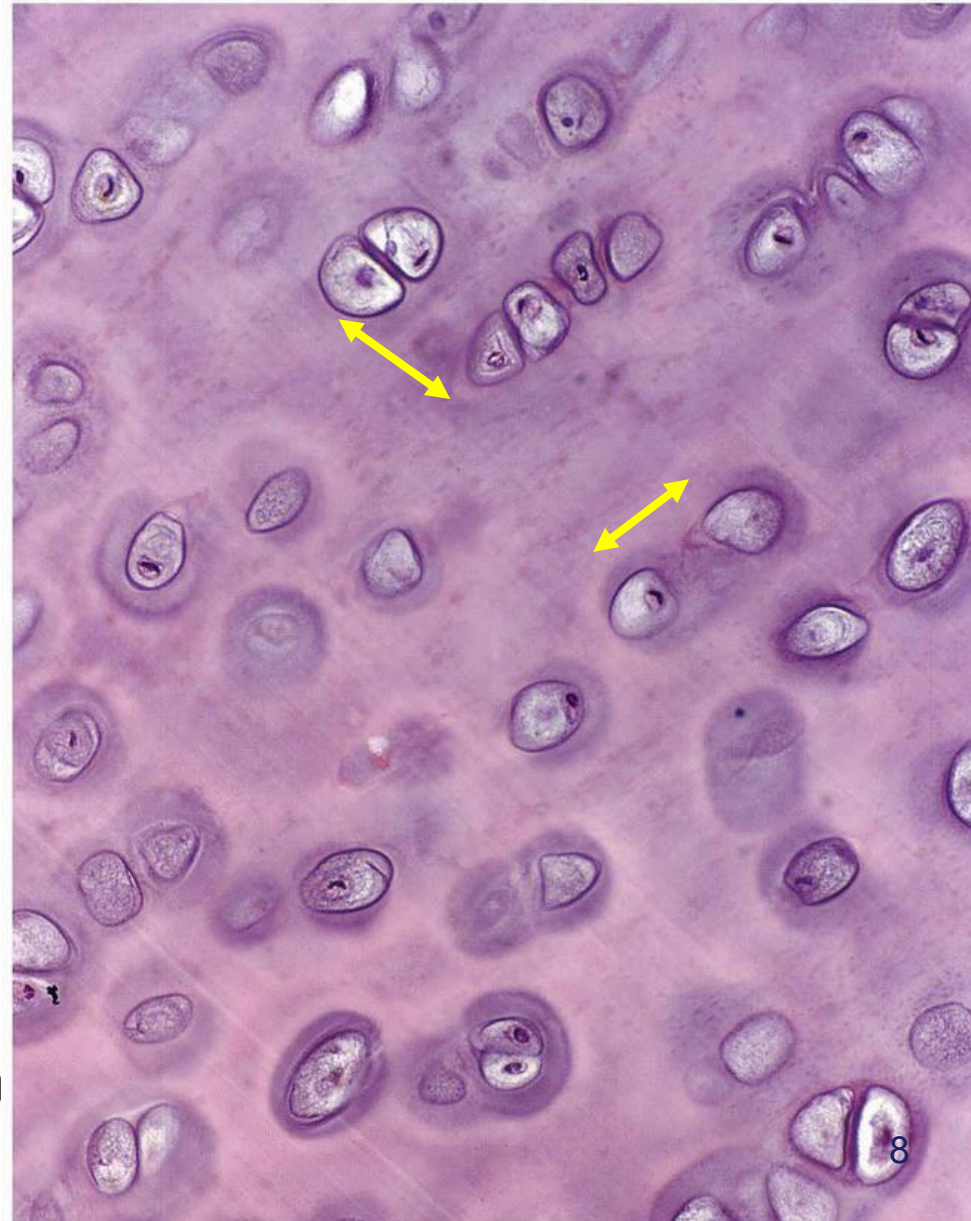
Cartilage cells

Chondrocytes

- As chondrocytes in isogenous groups mature, they move apart by their own synthesis of ECM. This results in the internal growth of cartilage.

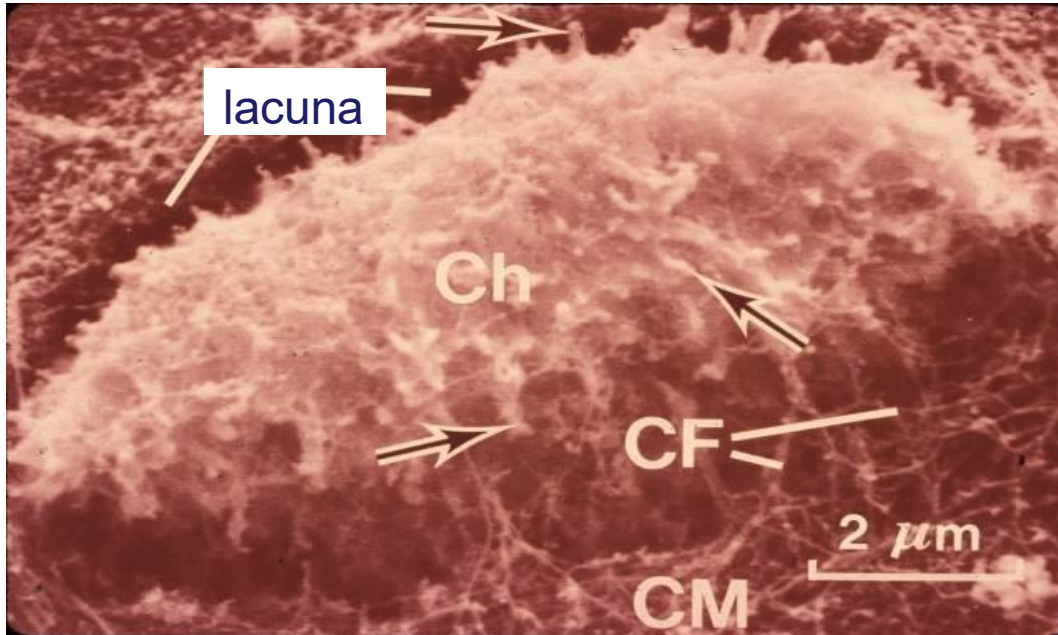


Cell nest or Isogenous group

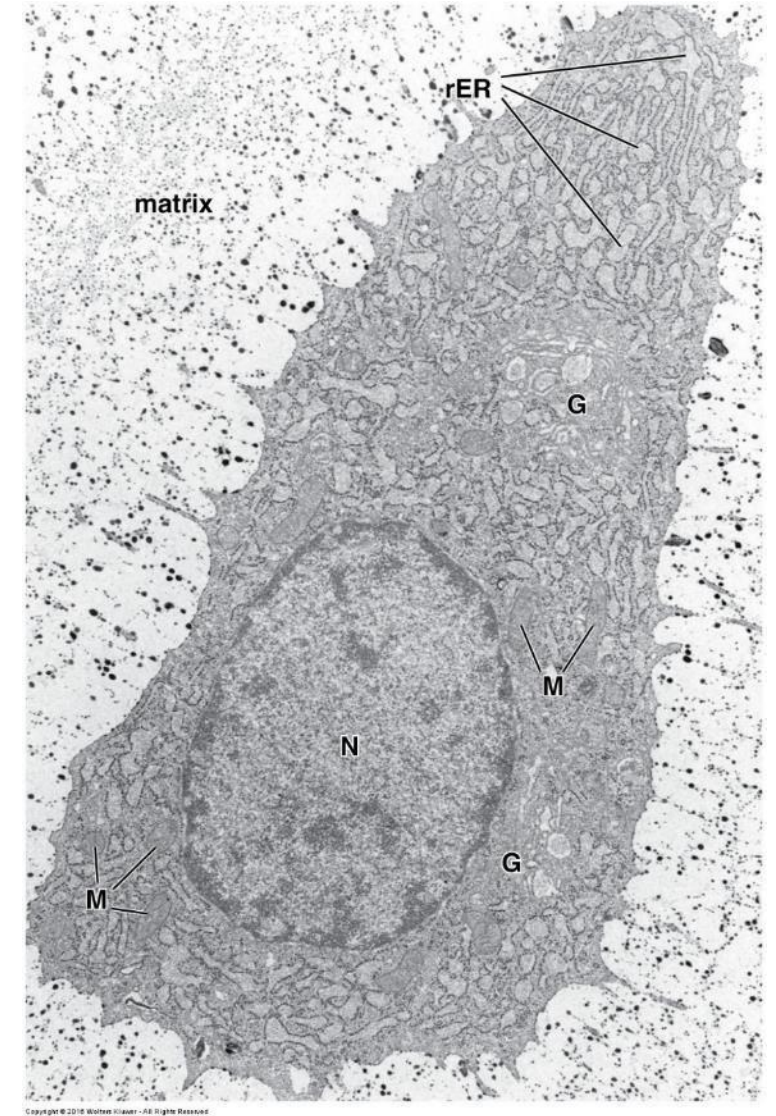


Electron micrographs:

- Irregular chondrocyte surfaces
- Extensive Golgi (G) apparatus
- Abundant rough ER
- Euchromatic nucleus
- Lipid droplets and glycogen deposits (in older, less active cells)



Scanning EM:
CH- chondrocyte
CF- collagen fibers
CM- cartilage matrix

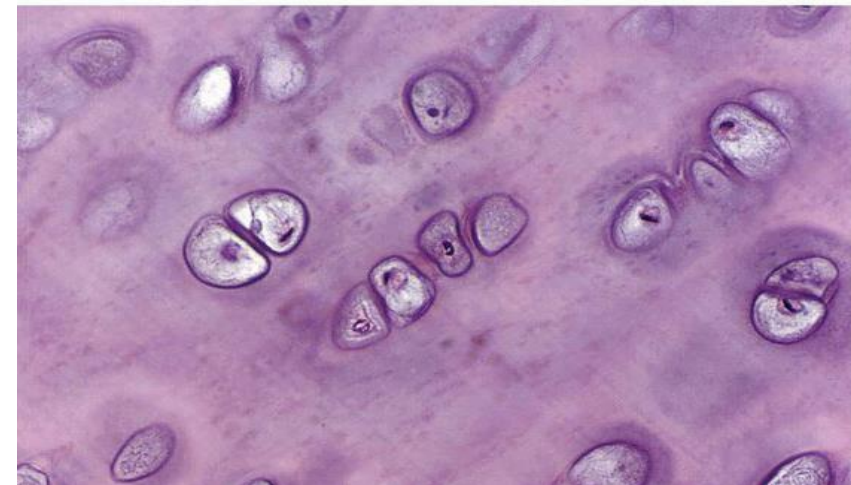
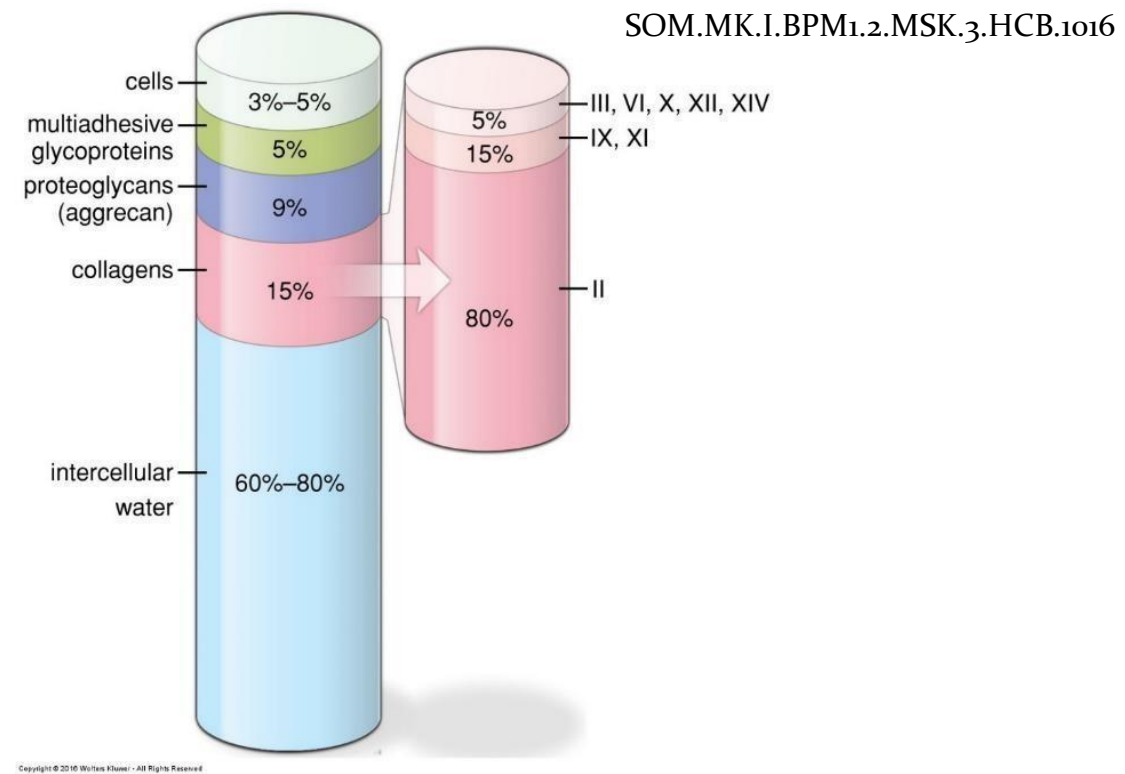


TEM of chondrocyte:

Extracellular Matrix (ECM)

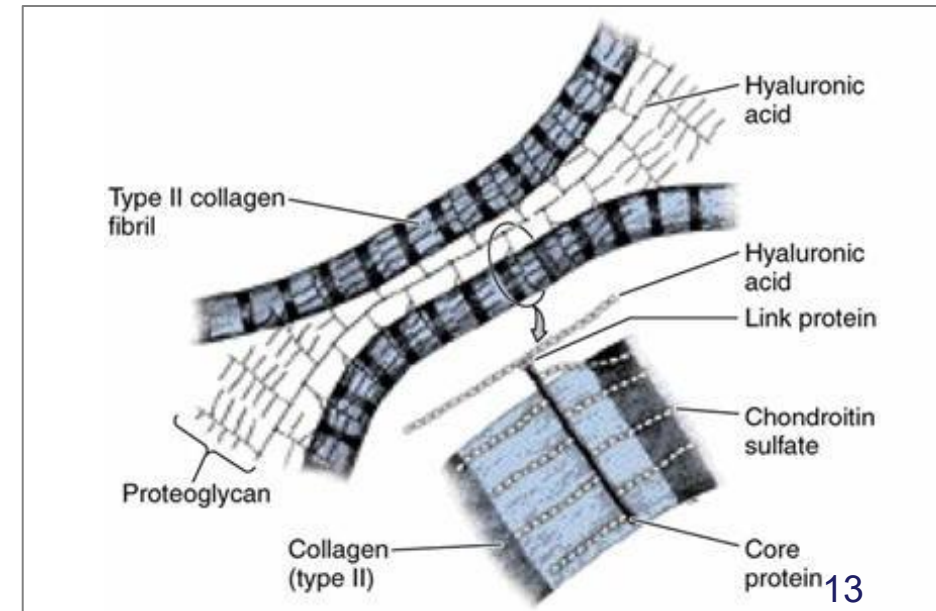
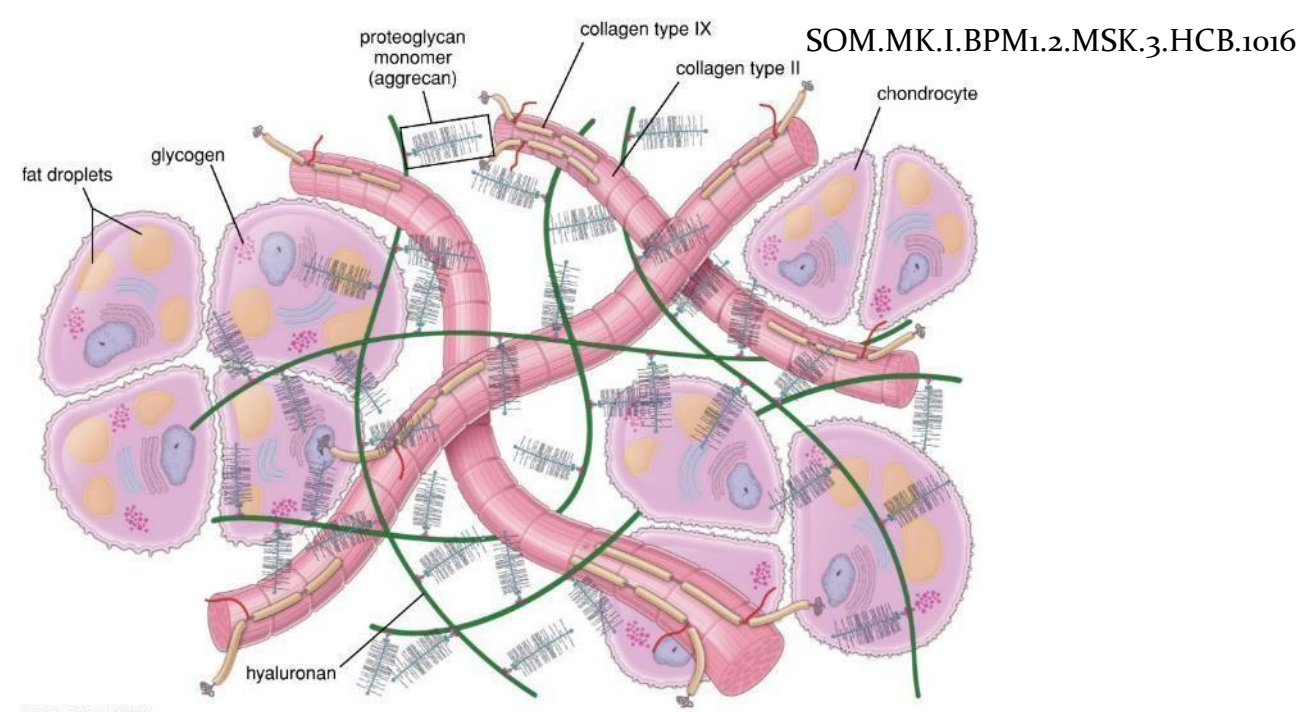
- Constitutes >95% of cartilage volume
- Composed of amorphous, hydrated ground substance and collagen fibrils
- Collagen constitutes about 15% of the total mass with **Type II collagen** being most abundant.
 - It is not readily discernible in histological sections because some of the fibrils (type II collagen) are very fine and have the same refractive index as the ground substance.

This gives hyaline cartilage matrix a “glassy” appearance.

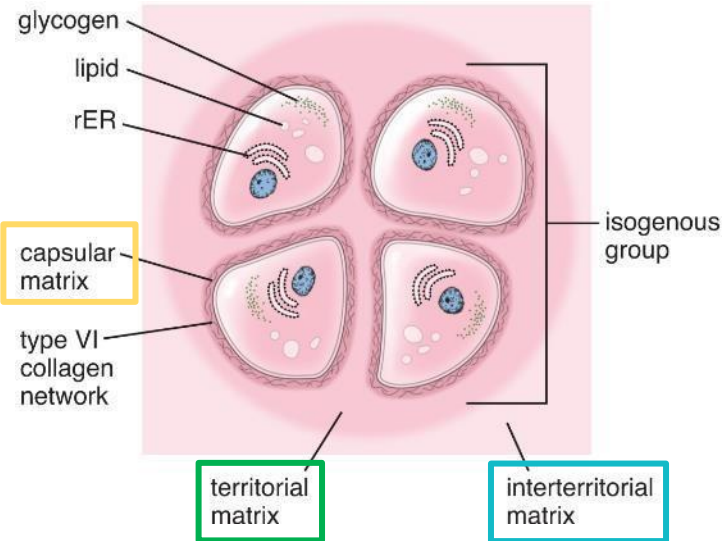


Composition of matrix:

- Sulphated groups present in **GAGs**, make it hydrophilic, enabling easy diffusion of nutrition to the cells
- Also provides resilience
- Presence of proteoglycans provide immense strength to the matrix so that cartilage can function as a model for bone formation.



Regions of cartilage Matrix



1. Capsular matrix

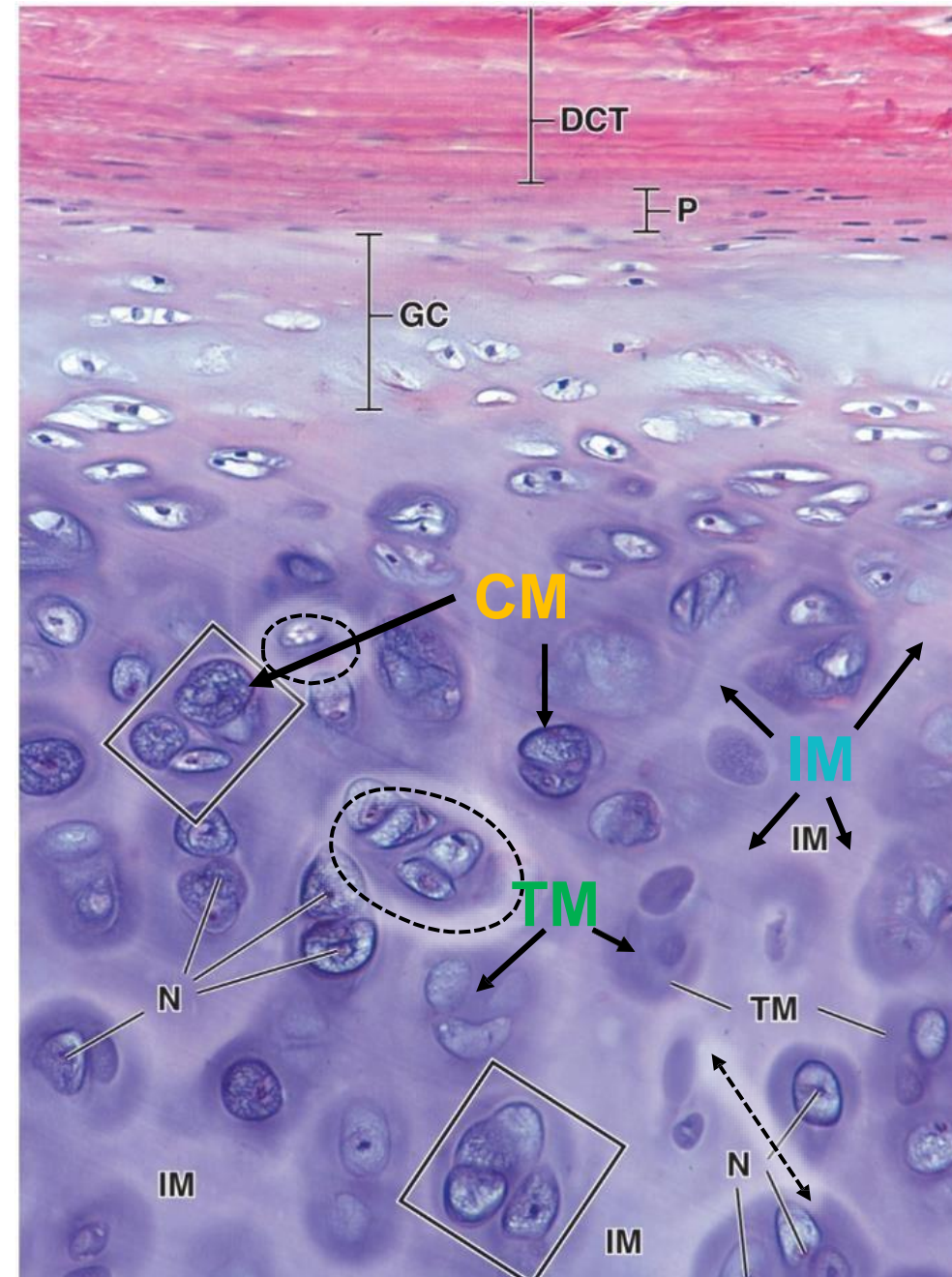
matrix adjacent to the chondrocyte
(also called pericellular matrix)

2. Territorial matrix (TM)

matrix found around and within an
isogenous group

3. Interterritorial matrix (IM)

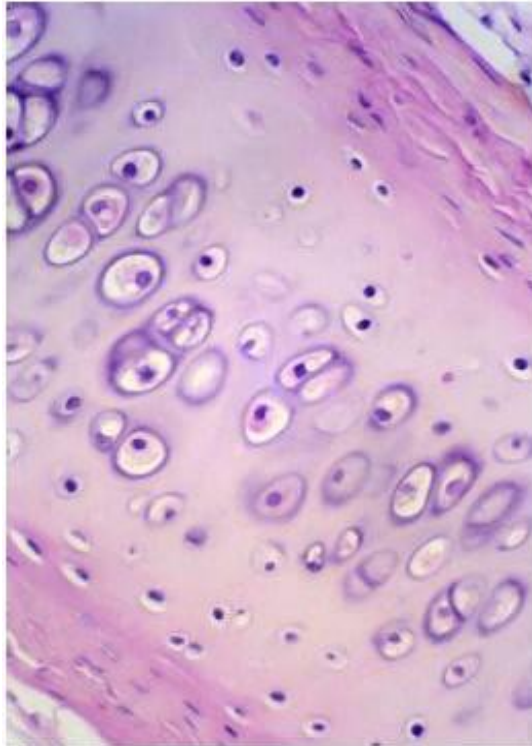
matrix in-between isogenous groups



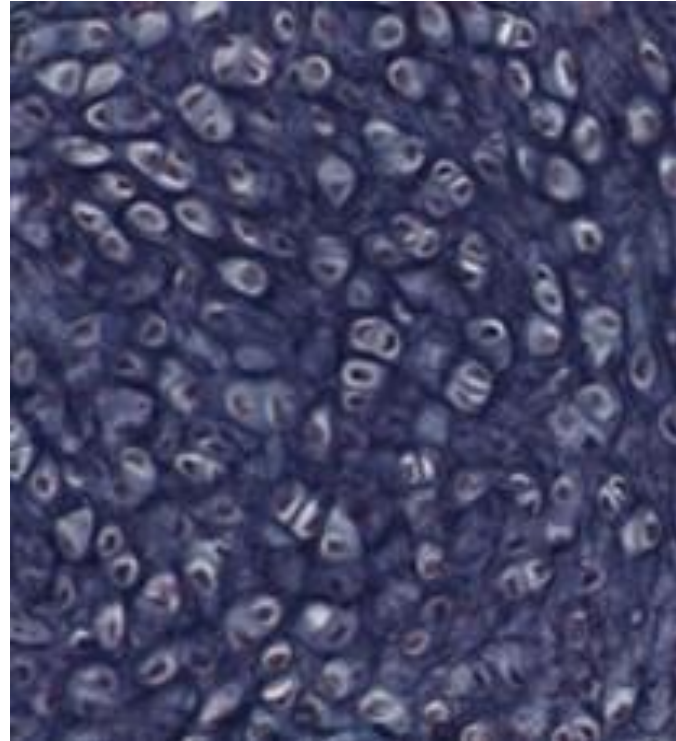
Regions of Cartilage Matrix	Composition
Capsular	• Highest concentration of sulfated proteoglycans, hyaluronan, biglycans, and several multiadhesive glycoproteins. • Less collagen fibers • Basophilic staining with H/E
Territorial	• Randomly arranged network of type II collagen fibrils with smaller quantities of type IX collagen. • Lower concentration of sulfated proteoglycans • Less basophilic than the capsular matrix.
Interterritorial	• Coarser collagen fibers. • Less intensely staining

Compare and contrast

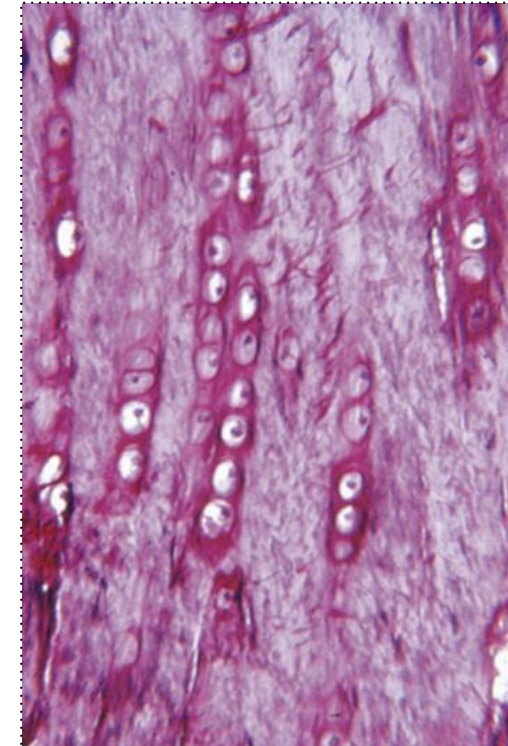
Hyaline Cartilage



Elastic Cartilage



Fibrocartilage



Make a table based on :

- a) locations, b) fibers in the ECM, c) presence or absence of perichondrium
- d) types of growth e) appearance of isogenous groups.

Ask yourself:

- What is cartilage?
- Name the 3 major types of cartilage and state 2 sites where each can be found.
- What type of collagen fibers are predominant in hyaline cartilage?
- What are the 2 layers of the perichondrium and what is the composition of each layer?
- How is fibrocartilage different from the other types?
- What are isogenous groups?
- What is the predominant ground substance found in cartilage?
- What are the 2 types of growths observed in hyaline cartilage?
- What is peculiar about articular surface of the long bones?
- When bone develops on cartilage model, what kind of bone formation is it? (...more on this in the 'Bone' lectures)





Clinical Cases

Use the information from the Cartilage Flipped Class objectives, the compulsory reference textbook reading assignment, and the preceding slides to prepare for these cases and related questions during the flipped class session. This is aimed at preparing you to engage fully in the flipped classroom session and facilitate and enhance your understanding of the material.

See you in class!

Cartilage Flipped Class

Additional Objectives

State the function(s) of hyaline cartilage.	SOM.MK.I.BPM1.2.MSK.3.HCB.1025
Describe the histological structure of hyaline cartilage.	SOM.MK.I.BPM1.2.MSK.3.HCB.1026
Give examples of the localizations of hyaline cartilage.	SOM.MK.I.BPM1.2.MSK.3.HCB.1027
State the function(s) of articular cartilage	SOM.MK.I.BPM1.2.MSK.3.HCB.1028
Describe the histological structure of articular cartilage.	SOM.MK.I.BPM1.2.MSK.3.HCB.1029
Give examples of locations of articular cartilages	SOM.MK.I.BPM1.2.MSK.3.HCB.1030
State the function of elastic cartilage.	SOM.MK.I.BPM1.2.MSK.3.HCB.1031
Describe the microanatomical structure of elastic cartilage.	SOM.MK.I.BPM1.2.MSK.3.HCB.1032
Give examples of locations of elastic cartilage.	SOM.MK.I.BPM1.2.MSK.3.HCB.1033
State the function of fibrocartilage.	SOM.MK.I.BPM1.2.MSK.3.HCB.1034
Describe the microanatomical structure of fibrocartilage.	SOM.MK.I.BPM1.2.MSK.3.HCB.1035
Give examples of locations of fibrocartilage.	SOM.MK.I.BPM1.2.MSK.3.HCB.1036
Discuss osteoarthritis and relate the histological changes in cartilage with clinical presentation.	SOM.MK.I.BPM1.2.MSK.3.HCB.1037

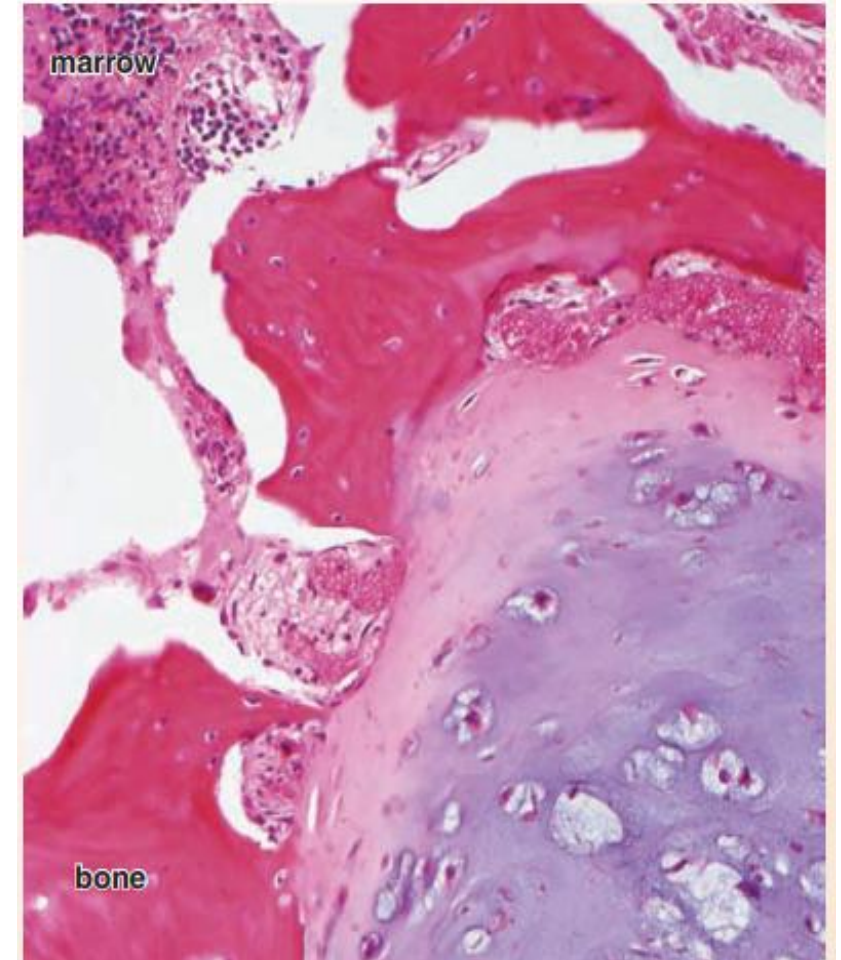
Clinical Case 1

A 62-year-old man comes to the physician complaining of worsening pain and pressure in his upper arm for the past 2 months. The patient states that he has felt dull pain in his right shoulder that gets worse with activity.

Physical examination shows tenderness and slight swelling of the proximal right humerus. Biopsy shows that the tumor is comprised of tissue similar to the tissue normally found within epiphyseal growth plate. The patient is diagnosed with a **chondrosarcoma**, a malignant tumor that secretes a cartilaginous matrix.

Which of the major types of cartilage typically constitutes the epiphyseal growth plate? Give examples of other locations in the body where we can find this tissue.

What are some of the histological characteristics of this tissue? How do these characteristics contribute to its functions?



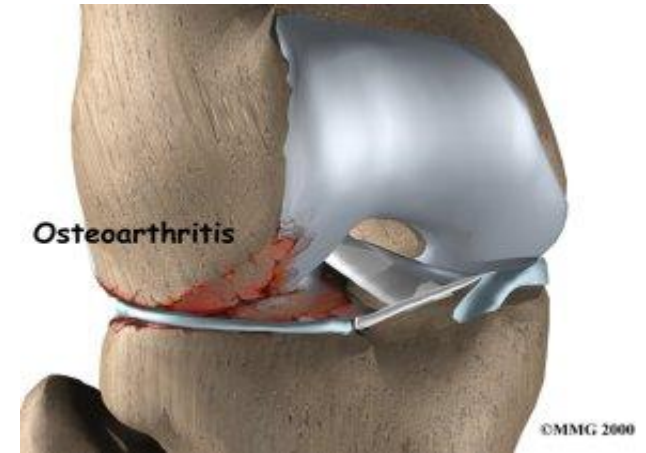
Clinical Case 2

A 58-year-old woman comes to the physician because of progressively worsening pain and stiffness in her knees for the past 6 months. She states the pain gets worse with physical activity and tends to improve with rest.

Physical examination shows tenderness on palpation of the medial aspect of her knees. Radiographic imaging shows bilateral joint space narrowing and osteophyte formation at the articular surfaces of the knee joints.

Describe the layers of the specialized tissue that normally lines the articular surfaces of long bones.

How do the cells of this tissue receive nutrients?



Clinical Case 3

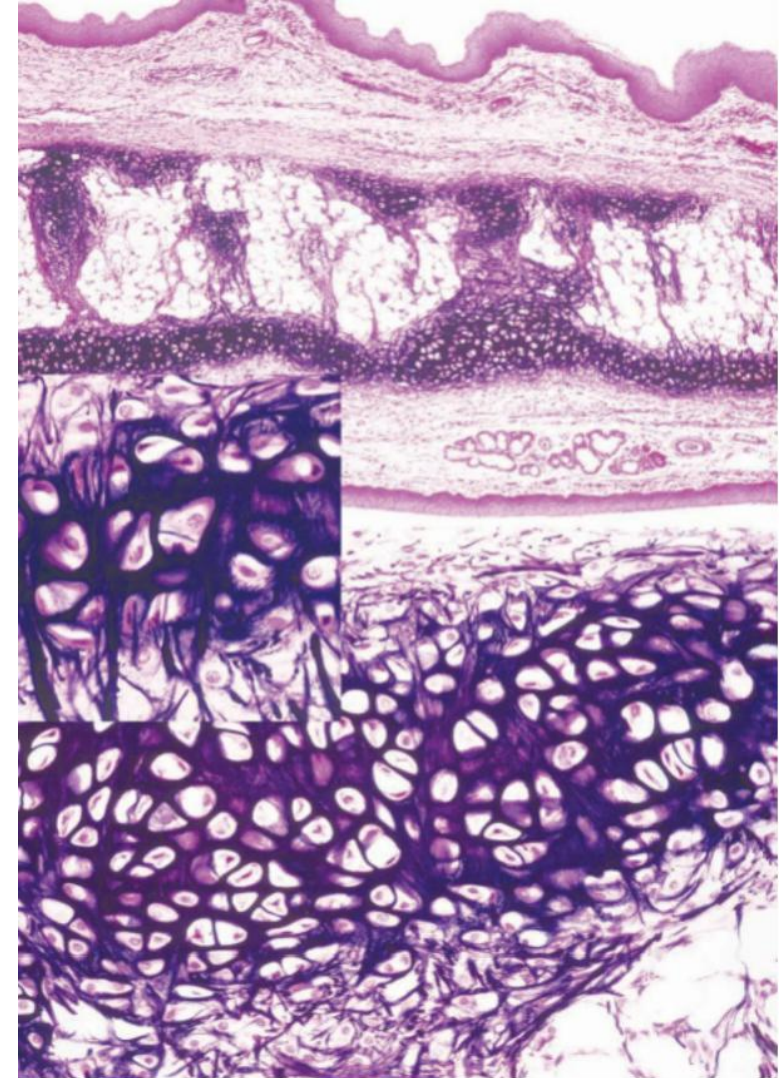
A 3-year-old boy is brought to the emergency department by his parents because of a 4-hour history of hoarseness, severe sore throat, and a high fever.

Physical examination shows an acutely ill child that is leaning forward with drool coming out of his mouth. Direct observation of the larynx reveals a swollen and erythematous epiglottitis.

He was diagnosed with **bacterial epiglottitis** and administer proper treatment and antibiotics.

What type of cartilage forms the core of the inflamed structure (epiglottis)?

How is this type of cartilage similar to hyaline cartilage? How are they different? (Compare composition, function, location, etc.).



Clinical Case 4

50-year-old woman comes to the emergency room because of severe lower back pain for the past 2 days. She reports the pain started after she attempted to pick up a heavy box at her workplace.

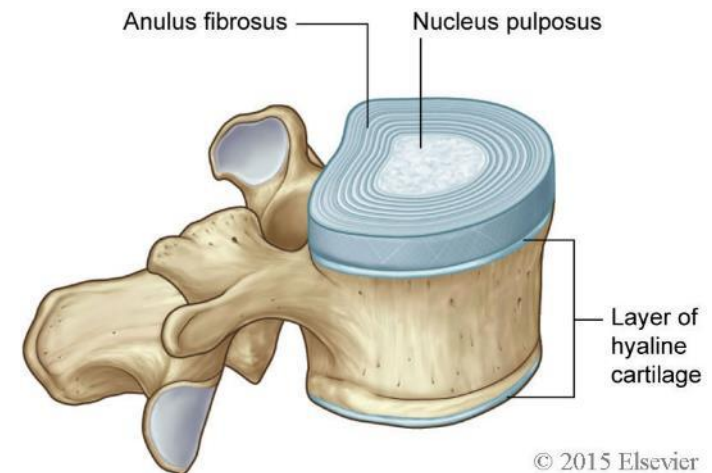
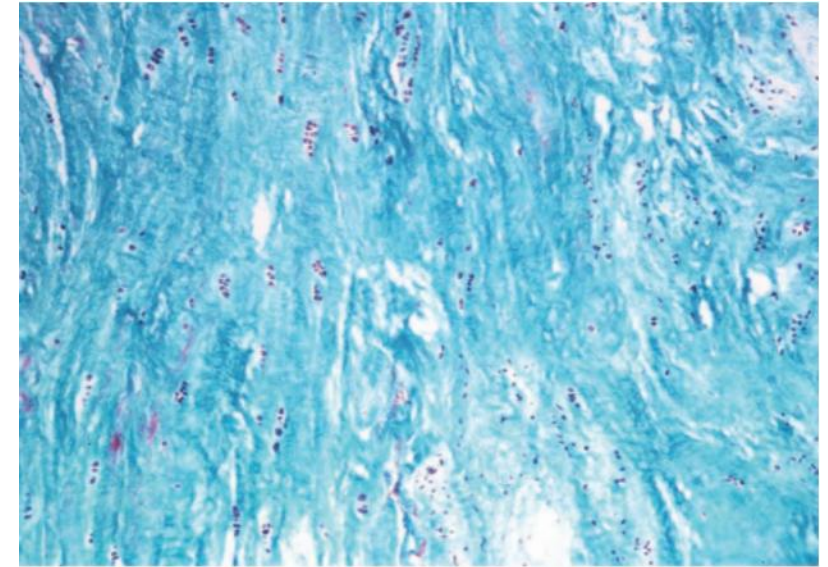
Physical examination shows tenderness in the lumbar region on movement and palpation. Imaging shows an intervertebral (IV) disc herniation at the L4 vertebral level.

What component of the IV disc was damaged to result in disc herniation?

What type of tissue is this damaged component made up of?

How does this tissue type differ from the other types of cartilage?

How does the development of this tissue explain its cellular composition?



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